

## Review

# Small-Scale Farming: A Review of Challenges and Potential Opportunities Offered by Technological Advancements

Rajveer Dhillon \* and Qianna Moncur

Agricultural Research Development Program (ARDP), Central State University, Wilberforce, OH 45384, USA; qmoncur.csu@centralstate.edu

\* Correspondence: rdhillon@centralstate.edu

**Abstract:** Smallholder farmers play an important role in ensuring world food security. In addition to food security, small-scale farming also provides numerous other direct and indirect environmental, social, cultural, and economic benefits by improving crop diversification, job security, and self-sufficiency. However, small-scale agriculture faces several challenges which are further exacerbated by climate change, population increase, water scarcity, and soil degradation. Agriculture has seen a lot of technological advancements in the last few decades. However, access to these advancements has not always been economically viable for small or medium farm operations. This article aimed to review the major barriers to small-scale farming and to review the status and potential opportunities offered by advanced technologies that can benefit small-scale holders. Based on the review, we found that economics, marketing, climate change, lack of awareness, educational resources, infrastructure, information, and technology are the major challenges to small-scale farming. Technologies such as unmanned aerial vehicles (UAVs), the IoT (internet of things), irrigation automation, and the use of smartphones are already becoming mainstream in small-scale farming and other technologies like robotics, artificial intelligence (AI), yield monitoring, and food traceability have potential opportunities to solve the challenges that hamper the success of small-scale growers.

**Keywords:** small-scale farming; smallholders; food security; technology; agriculture technology; precision agriculture; ag-tech



**Citation:** Dhillon, R.; Moncur, Q. Small-Scale Farming: A Review of Challenges and Potential Opportunities Offered by Technological Advancements. *Sustainability* **2023**, *15*, 15478. <https://doi.org/10.3390/su152115478>

Received: 22 August 2023

Revised: 24 October 2023

Accepted: 26 October 2023

Published: 31 October 2023



**Copyright:** © 2023 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

## 1. Introduction

Undoubtedly, smallholder farmers play a very important role in the world's food security by developing sustainable food production systems for many communities and countries. It should be noted that, based on income, the scale for smallholder farmers is different in different parts of the world. For this article, as per the United States Department of Agriculture (USDA) a farmer who is considered a smallholder farmer is one who has an operation with gross cash farm income under \$250,000. More than one-third of the world's food is produced by smallholder farming systems from 24% of gross agricultural land [1]. Most of these small farming systems are the primary source of food for the population in developing countries [2]. Similar scenarios can be found in developed countries as well. According to the 2012 census of agriculture from the United States Department of Agriculture, small family farms in the US operated 48 percent of all farmlands, accounted for one-fifth of agriculture sales, but only made up five percent of US net farm income [3]. More than 90% of total farms fall in the category of small farms in many of the states in the US. The success and upliftment of smallholder farming systems will play a key role in feeding the Earth's predicted nine billion population by 2050. Small farmers are crucial to the socioeconomic advancement and political stability of rural areas because they produce goods that help locals maintain their food security, nutrition, and health. A rapid increase in population may increase demand for food goods in general, and products from small farmers especially [4]. High yields and productivity with a significantly

decreased environmental imprint are required for a sustainable food-secure future. This cannot be achieved without the multitude of small-scale farmers [5]. The authors of [5] conducted a 10-year study assessing the use of adopted technology providing high yields and low pollution. It was determined that providing small-scale farmers with improved management technology to assist them in achieving higher productivity and environmental performance is an attempt at a world with a sustainable future. Small-scale farming throughout the world provides numerous other direct and indirect environmental, social, cultural, and economic benefits by improving crop diversification, job security, and self-sufficiency. However, small-scale agriculture is facing several challenges like lower income, lack of technology, poor access to information, market and certification barriers, and labor shortages [6]. In addition, these challenges are exacerbated by climate change, population increase, water scarcity, and soil degradation.

Agriculture, specifically large-scale agriculture, has seen a lot of technological advancements in the last few decades. This substantial transformation in agriculture was led by advancements in precision agriculture (PA) with technologies like GNSS, auto-steering, crop and soil sensors, rate control technologies of crop inputs, and yield mapping [7]. PA technologies have proven their worth in many ways, for example, (a) by making farm operations automated, more efficient, and sustainable, (b) by enhancing agriculture profitability and productivity, and (c) by providing environmental benefits with reduced use of chemicals and natural resources. Precision farming adopters from Midwestern states in the US have reported that even though there is scope for improvement, these technologies already help them in many ways [8]. However, PA has not always been an economically viable option for small or medium farm operations [9]. The aim of this article is two-fold: (a) the scientific aim of this article is to review the literature to summarize and categorize the major challenges faced by smallholder farmers, and (b) the utilitarian aim of this article is to provide a list and current status of the use of modern technologies that have the potential to benefit smallholder farmers. The overarching goal of this article is to highlight the importance of small-scale farming, and its challenges, and to list the recent technological advancements that can benefit small-scale farmers to achieve success, which is eventually very vital for the sustainability of food production, especially for rural areas and developing countries.

## 2. Major Barriers Faced by Small-Scale Growers

Even though there are many benefits of small-scale farming, there are multiple barriers that need to be tackled to help make small-scale farming profitable and sustainable. As is the case with small-scale farming in the USA, similar challenges have been faced by small-scale farmers all over the world including places like India, Ukraine, the United Kingdom, and a few other African and Asian countries. Even though farmers have been faced with some unique challenges in developing countries due to arable land and water supply shortages, it can be very helpful to be aware of challenges faced worldwide as small-scale farmers in many countries are experiencing similar if not the same issues. There are a wide variety of challenges in global small-scale farming, some are related to economics, marketing, labor costs, lack of resources and technology, decline in production, climate change, lack of education, and insufficient infrastructure. Based on our review of the literature, the following are the main categories for the challenges faced by small-scale growers. These articles are also presented in table format in Table 1.

**Table 1.** Major barriers faced by small-scale farmers.

| Challenge | Article                                                                                                                    | Authors                         | Country        | Summary/Results                                                                                                                                                                                                                                                                                                |
|-----------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Economic  | small-scale produce growers' barriers and motivators to value-added business: Food safety and beyond                       | Chen et al., 2021 [10]          | USA            | Economic challenges are one of the major barriers to the small-scale grower's ability to have a successful farming business. It has been found that labor is a limiting factor when it comes to starting a small-scale farming business and it also limits farmers' ability to scale their business.           |
|           | Constraints and challenges facing the small-scale farmers in Limpopo Province, South Africa                                | Mpandeli et al., 2014 [11]      | South Africa   | Challenges with the cost of transportation and price of inputs such as fertilizers and herbicides have risen in the Limpopo Province of South Africa, hindering agricultural development in this province and others in South Africa where they are experiencing poverty.                                      |
|           | Learning from the South: common challenges and solutions for small-scale farming                                           | Stringer et al., 2008 [12]      | United Kingdom | A workshop has been held in the United Kingdom to identify the challenges common to small-scale farmers and possible solutions for them. As a result, economic challenges were identified, and it has been concluded that giving farmers a safe space to discuss their challenges has proven to be successful. |
|           | Small Farmers, Big Challenges: A Needs Assessment of Florida Small-Scale Farmers' Production Challenges and Training Needs | Goodwin et al., 2013 [13]       | USA            | Studies took place to assess challenges with small-scale farmers in Florida. Economic challenges were found to be one of the major barriers among several others within the population.                                                                                                                        |
|           | An investigation into the use of ICT in the provision of agricultural information to small-scale farmers in Harare         | Chisita et al., 2010 [14]       | Zimbabwe       | Zimbabwe is an agricultural country due to most of the population being based in rural areas. Some barriers being faced by small-scale farmers are access, efficiency, and affordability pertaining to agricultural information.                                                                               |
|           | Structural Changes in U.S. Agriculture: Implications for Small Farms                                                       | Gebremedhin et al., 1996 [15]   | USA            | An analysis took place to observe the problems faced by small farms. This study concluded that the shifting structure of agricultural production and economic challenges were strong factors.                                                                                                                  |
|           | The perception of and adaptation to climate variability/change in Ghana by small-scale and commercial farmers              | Yaro et al., 2013 [16]          | Ghana          | Qualitative interviews were conducted to assess the challenges experienced by small farmers. It was found that the impact of climate change is more destructive for farmers with economic problems and small-scale farmers.                                                                                    |
| Marketing | Small Farmers, Big Challenges: A Needs Assessment of Florida Small-Scale Farmers' Production Challenges and Training Needs | Goodwin et al., 2013 [13]       | USA            | In Florida, 6 focus groups were conducted to discuss the challenges of small-scale farmers. One of the themes that stemmed from the group discussions was marketing, and it was mostly because of the lack of funds for advertising.                                                                           |
|           | Challenges of Small-Scale Farming In Ukraine                                                                               | Bezus et al., 2019 [17]         | Ukraine        | A study that took place identified two main challenges faced by Ukrainian farmers. The challenges were in marketing and the production process.                                                                                                                                                                |
|           | Exploring Challenges and Opportunities Embedded in Small-scale Farming in Zimbabwe                                         | Kang'ethe and Serima, 2014 [18] | Zimbabwe       | Small-scale farming has been deemed very important since the early 1980s. However, challenges were faced by small-scale farmers in Zimbabwe. A few of the challenges were marketing, land contracting, and weaker government policies.                                                                         |

Table 1. Cont.

| Challenge      | Article                                                                                                                                                                 | Authors                   | Country          | Summary/Results                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Marketing      | Linking small fruit and vegetable farmers and institutional food service operations: marketing challenges and considerations                                            | Boys et al., 2019 [19]    | USA              | Studies took place in the USA to determine the marketing challenges of small-scale fruit and vegetable farmers. Poor infrastructure, prices and payment terms, and insurance and certification requirements and barriers were identified.                                                                                                                                                                                              |
|                | Market Access and Sunflower Marketing: Challenges and Prospects to Small scale farmers in Tanzania                                                                      | Ugulumu et al., 2014 [20] | Tanzania         | After utilizing data from a sample, findings presented that market access hinders small growers from accessing markets, internationally and domestically.                                                                                                                                                                                                                                                                              |
| Climate change | Technology for Small Scale Farmers in Tanzania: A Design Science Research Approach                                                                                      | Misaki et al., 2016 [21]  | Tanzania         | Small-scale farmers in Tanzania are faced with many challenges, some are lack of infrastructure, food insecurity, changing climate conditions, and the lack of access to credit and services. The agriculture sector in Tanzania remains poor despite multiple initiatives.                                                                                                                                                            |
|                | Factors impacting sugarcane production by small-scale farmers in KwaZulu-Natal Province-South Africa                                                                    | Thibane et al., 2023 [22] | South Africa     | Smallholder sugarcane production is a vital sector in many countries, including South Africa. Unfortunately, there have been challenges that have contributed to the decline of smallholder sugarcane production. A study took place to identify factors contributing to the decline of sugarcane production. Results showed that drought stress, labor costs, and lack of finances contribute to the decline of sugarcane production. |
|                | Challenges for sustainable maize production of smallholder farmers in sub-Saharan Africa                                                                                | Cairns et al., 2021 [23]  | Africa           | Four challenges were found for maize production in Africa: maize nutrient density, varietal replacement, fertilizer usage, and gender gaps.                                                                                                                                                                                                                                                                                            |
|                | Limited access and use of climate information by small-scale sugarcane farmers in South Africa: A case study                                                            | Ncoyini et al., 2022 [24] | South Africa     | Small-scale growers have been experiencing significant challenges in growing sugar cane in South Africa. This study concluded that these farmers lack access to climate information.                                                                                                                                                                                                                                                   |
|                | Environmental Challenges Threatening the Growth of Urban Agriculture in the United States                                                                               | Sam and Lovell, 2013 [25] | USA              | Challenges such as soil, water, and atmospheric pollution, unfavorable climatic conditions, and safety have emerged for urban farmers.                                                                                                                                                                                                                                                                                                 |
|                | Climate Change and Small Farmers' Vulnerability to Food Insecurity in Cameroon                                                                                          | Mbuli et al., 2021 [26]   | Cameroon, Africa | Although the small-scale farmers in this scenario have a resource-rich economy, they are vulnerable to climate change. With the increase in high temperatures globally, increasing precipitation, and severity of storms, climate change is threatening food security.                                                                                                                                                                 |
|                | Assuring food security in developing countries under the challenges of climate change: key trade and development issues of a fundamental transformation of agriculture. | Hoffman et al., 2011 [27] | Switzerland      | Climate change can potentially damage natural resources that agriculture depends on. Sustainable production systems are required that improve the productivity of small-scale farmers.                                                                                                                                                                                                                                                 |

Table 1. Cont.

| Challenge                                                      | Article                                                                                                                                                   | Authors                      | Country              | Summary/Results                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lack of:<br>Aware-<br>ness,<br>educational<br>resources        | Food Safety Education Needs Assessment for Small-Scale Produce Growers Interested in Value-Added Food Production                                          | Chen et al., 2022 [28]       | USA                  | Small-scale produce growers have taken an interest in value-added food production. Observations reported that small-scale produce growers lack enough food safety awareness.                                                                                                                                                                                               |
|                                                                | Organic Farming and Small-Scale Farmers: Main Opportunities and Challenges                                                                                | Jouzi et al., 2017 [6]       | Developing Countries | Due to arable land and water supplies being reduced globally, farmers have been faced with challenges in developing countries. Lower yields compared to traditional systems, issues with soil nutrient management, certification and market restrictions, and small-holder teaching and research requirements are among the key challenges of this food production system. |
|                                                                | Assessment of post-harvest challenges of small-scale farm holders of tomatoes, bell, and hot pepper in some local government areas of Kano State, Nigeria | Olayemi et al., 2010 [29]    | Nigeria              | A survey was conducted to assess the challenges associated with small-scale farmers growing tomatoes, bell peppers, and hot peppers in Nigeria. The survey concluded that poor management and little knowledge of farming contributed to harvest decline.                                                                                                                  |
|                                                                | Making Small-Scale Farming Work in Sub-Saharan Africa                                                                                                     | Afenyo et al., 2012 [30]     | Africa               | Challenges resulted in poor performance and lack of improvement for small-scale farmers in Africa. Farmers also lack new skills that are needed for the evolving environment.                                                                                                                                                                                              |
|                                                                | Constraints and challenges facing the small scale farmers in Limpopo Province, South Africa                                                               | Mpandeli et al., 2014 [11]   | South Africa         | Challenges with the cost of transportation and price of inputs such as fertilizers and herbicides have risen in the Limpopo Province of South Africa, hindering agricultural development in this province and others in South Africa where they are experiencing poverty.                                                                                                  |
|                                                                | Utilisation of mobile phones in accessing agricultural information by smallholder farmers in Dzindi irrigation scheme in South Africa                     | Mavhunduse et al., 2019 [31] | South Africa         | Challenges arose among small-scale farmers in rural areas. Studies took place to examine the use of mobile phones to provide information. Farmers need help with utilizing the smart features on their phones.                                                                                                                                                             |
|                                                                | Social grant dependence, irrigation water use and on-farm entrepreneurial spirit: a behavioural explanation for smallholders in KwaZulu-Natal.            | Zaca 2018 [32]               | South Africa         | Farmers in South Africa have been facing challenges with finding relevant information for running farming businesses.                                                                                                                                                                                                                                                      |
| Lack of:<br>Infrastruc-<br>ture,<br>information,<br>technology | Access to and Utilization of Information and Communication Technologies by Agricultural Researchers and Extension Workers in Zimbabwe                     | Mugwisi et al., 2015 [33]    | Zimbabwe             | Sometimes improper utilization can be an issue. This study shows inadequate use of resources by extension workers and researchers that can improve the agriculture production of poor rural families.                                                                                                                                                                      |
|                                                                | Challenges facing sub-Saharan small-scale farmers in accessing farming information through mobile phones: A systematic literature review                  | Misaki et al., 2018 [34]     | Tanzania, Africa     | Small-scale farmers located in sub-Saharan Africa are faced with challenges regarding mobile phone technology. Because of these challenges, they are faced with the difficulty of gaining reliable information. The low education level and training of these small-scale farmers becomes a barrier when using mobile phones, they are also costly.                        |
|                                                                | An Overview of Internet of Things (IoT) and Data Analytics in Agriculture: Benefits and Challenges                                                        | Elijah et al., 2018 [35]     | Ghana                | In Ghana, farmers face scarce natural resources, limited availability of land, and more. Studies took place to find the benefits of IoT.                                                                                                                                                                                                                                   |



Table 1. Cont.

| Challenge                                                 | Article                                                                                                                                                        | Authors                    | Country      | Summary/Results                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lack of:<br>Infrastructure,<br>information,<br>technology | Regulatory and Other Barriers to Urban and Peri-urban Agriculture: A Case Study of Urban Planners and Urban Farmers from the Greater Chicago Metropolitan Area | Castillo et al., 2013 [36] | USA          | A few barriers have been recorded in Chicago with urban farmers. Some of the barriers being faced are limited access to land, insufficient infrastructure, lack of research, and lack of skills in urban agriculture.                                                                      |
|                                                           | Small Farmers, Big Challenges: A Needs Assessment of Florida Small-Scale Farmers' Production Challenges and Training Needs                                     | Goodwin et al., 2013 [13]  | USA          | Studies took place to assess challenges with small-scale farmers in Florida. challenges accessing new information were also found to be significant within the population.                                                                                                                 |
|                                                           | The challenges of utilizing information communication technologies (ICTs) for the small-scale farmers in Zambia                                                | Kalusopa, 2005 [37]        | Zambia       | The application of information communication technologies (ICT) was found to be a challenge.                                                                                                                                                                                               |
|                                                           | Challenges and Opportunities of Information and Communication Technologies for Dissemination of Agricultural Information in Ethiopia                           | Getahun, 2020 [38]         | Ethiopia     | Inefficiency and cost of technology was found to be a major barrier.                                                                                                                                                                                                                       |
|                                                           | Moving local food through conventional food system infrastructure: value chain framework comparisons and insights                                              | Bloom et al., 2010 [39]    | USA          | It is considered that direct marketing of produce by locals often cannot meet rapidly expanding consumer demand. This study shows there is a need for transitional food system infrastructure that benefits direct marketing.                                                              |
|                                                           | Patterns of Access To Rural Service Infrastructure: The Case of Farming Households In Limpopo Province                                                         | Makhura et al., 2003 [40]  | South Africa | This study used factor analysis to find access to infrastructure in rural communities and the results show the concept of a center-periphery pattern of service infrastructure. Infrastructure development policy and programs need to consider rural economic development.                |
|                                                           | Agricultural cooperatives II: Can they facilitate access of small-scale farmers in South Africa to input and product markets?                                  | Ortman et al., 2007 [41]   | South Africa | This study shows how low levels of education and literacy, lack of market information, insecure property rights, poor road and communication infrastructure, and lengthy distances to markets contribute to significant transaction costs for farmers in two KwaZulu-Natal communal areas. |

### 2.1. Economic Challenges

Economic challenges are one of the major barriers to the small-scale grower's ability to have a successful farming business. Small-scale farmers are faced with many economic challenges, for example, labor costs, lower income, and high crop-input cost to name a few. It has been found that labor is a limiting factor when it comes to starting a small-scale farming business and it also limits farmers' ability to scale their business [10]. An analysis took place to observe the problems faced by small farms and the study concluded that economic challenges were a strong factor and it also concluded that the shifting structure of agriculture production has important implications on the survival of small farms [15]. In fact, farmers in Durban, South Africa, have been struggling with the cost of urban farming because large-scale farming is on the rise, on the other side small-scale farmers have low incomes and most of them are living below the poverty line [42]. Challenges with transportation costs and crop-input costs have risen in the Limpopo Province of South Africa, hindering agricultural development in this province and others in South Africa, where they are experiencing poverty [11]. The economy of Durban, South Africa, prevents farmers from successfully pursuing urban farming. The cost is unmanageable on their low incomes [42]. Focus groups were conducted to identify the cause of this barrier throughout Florida and, following the study, it was shown that economic challenges were

one of the six themes included. The issue was about affording expenses to comply with regulations. The fluctuation of prices in the economy has affected their farming operations, and the small farmers also lack proper funding [13]. This entails adapting to changes in the market, globalization, government policies, and poverty. A workshop has been held in the United Kingdom to identify the common challenges between small-scale farmers and possible solutions for them. As a result, economic challenges were identified, and it has been concluded that giving farmers a safe space to discuss their challenges has proven to be successful [12]. Economic challenges are also intertwined with other challenges; according to a study it was found that the impact of climate change is more destructive for farmers with economic problems and small-scale farmers [16].

## 2.2. Marketing Challenges

Selling crops at the right price that maximizes the profitability of farming businesses and at the right time before crop or produce quality starts to deteriorate is another major challenge for small-scale farmers internationally. The time needed to market farm products has also been a barrier for these farmers due to the cost associated with advertising; it has become expensive. Another study took place in the USA to determine the marketing challenges of small-scale fruit and vegetable farmers. Poor infrastructure, price and payment terms, insurance, and certification requirements were the main barriers identified [19]. In Florida, USA, six focus groups were conducted to discuss the challenges of small-scale farmers. One of the themes that stemmed from the group discussions was marketing, and it was mostly because of the lack of funds for advertising. Small-scale farmers do not have as much access compared to the larger farmers. Another factor was international competition. The small-scale farmers felt pressured by the government to adhere to food safety regulations, yet another country could export their products without following these regulations [13]. Another study mentioned the challenges for sunflower farmers, the findings presented that market access hinders small growers from accessing markets, international and domestic [20]. According to the survey the farmers considered the need for market search for their products one of the most important factors. Small-scale farmers also mentioned that military conflict has caused them to lose their trade links, leading them to enter new markets and make some changes in their marketing strategies [17].

## 2.3. Climate Change

The effect of climate change on production is another major concern for small-scale farmers. Climate change has taken a toll on small-scale farmers because they rely heavily on crop-friendly weather for production and rainfall for irrigation. With temperatures rising annually, precipitation levels have been fluctuating, causing drought or flood situations. Smallholder sugarcane production is a vital sector in many countries. Unfortunately, there have been challenges that have contributed to the decline of smallholder sugarcane production. A study found that climate change can potentially damage natural resources that agriculture depends on; therefore, sustainable production systems are required that improve the productivity of small-scale farmers [27]. Another study took place to identify factors contributing to the decline of sugarcane production. Results showed that drought stress due to climate change is one of the primary reasons for the decline of sugarcane production [22]. Small farmers were interviewed in Kenya to discuss the decline of their maize production levels and it was found that climate change has been a factor in the decline of maize production [23]. The agriculture sector remains poor despite multiple initiatives in this sector [21]. South Africa continues to experience an irregular climate. The farmers experience significant challenges in growing sugar cane. Another study analyzed that these farmers lack access to climate information [24]. Small and socially disadvantaged farmers lack access to climate information relevant to their crops, which is one of the major barriers to effective climate change adaptation, and this information should be made available to small farmers through joint efforts from government agencies, international organizations, researchers and academics, community-based organizations,

and non-governmental organizations [43]. According to the survey [43], sometimes the reason for not having access to climate information could be the age or gender of the household and a lack of access to communication devices (radio, television, or phone). Data was collected to identify factors that had an impact on urban agriculture. Results show that there has been an unreliable source of water between the three settlements in Durban, not only for farming, but daily use, and they would have to pay for it. The poor quality of water from local rivers and streams has caused these farmers not to rely on that source for farming. The lack of land availability and poor soil quality became a barrier for them as well. To aid in combating these barriers, initiatives and programs need to take place and focus on water-efficient solutions to help these communities [42]. Crop yield has been declining in India, despite the utilization of chemical products. Pesticides and fertilizers have caused harm to agriculture. Organic farming is more beneficial to the environment. It will help with climate change and protect the ecological environment [44]. A study found that even small-scale farmers in resource-rich economies are vulnerable to climate change. With the increase in high temperatures globally, increasing precipitation and the severity of storms, climate change is threatening food security [26].

#### *2.4. Lack of Awareness and Educational Resources*

Small-scale growers have taken an interest in value-added food production. However, some studies have reported that small-scale produce growers in the USA lack food safety awareness [28]. They also lack the new skills that are needed for the evolving environment [30]. According to a survey that took place to assess food safety perceptions and to determine the gap in vendor and management food safety knowledge, as well as to find out the challenges and causes that prohibit important food safety measures from being implemented, the primary barriers to implementing food safety procedures were a lack of facilities, equipment, and materials including food safety rules particular to farmers markets. According to the findings, the primary barriers to implementing food safety procedures were a lack of facilities, equipment, and materials including food safety rules particular to farmers markets [45]. Another survey was conducted to assess the challenges associated with small-scale farmers growing tomatoes, bell peppers, and hot peppers in Nigeria. The survey concluded that poor management and little knowledge of farming contributed to harvest decline [29]. Lower yields, lack of knowledge regarding soil nutrient management, certification, and market restrictions, and a lack of small-holder-focused research and extension programs are among the key challenges of this food production system [6]. A study took place to assess these barriers in Limpopo Province, South Africa. The results show that these challenges have prevented agricultural development in this Province. Despite 30% of farmers having access to irrigation water, the misuse and poor management of water resulted in shortages. As for market access, even if production was high for them, their products would be wasted due to the scarce market. It was concluded that many of their issues can be resolved with the help of certain agricultural organizations and training programs [11]. Smallholders located in Africa are faced with challenges regarding mobile phone technology. A study took place to examine the use of mobile phones to provide information to farmers; however, it was found that farmers need help with utilizing the smart features on their phones [31]. Sometimes improper utilization of technology can be an issue. An article shows inadequate use of resources by extension workers and researchers, whereas adequate use of these resources can improve the agriculture production of poor rural families [33]. Another study took place to identify challenges that take place using mobile phones. As a result, several issues were found, i.e., lack of trust, poor education, high cost, inadequate infrastructure, theft, and more. Because of these challenges, farmers are faced with the difficulty of gaining reliable information. The low education level and training of these small-scale farmers becomes a barrier when using mobile phones, and they are also costly. The lack of training can be solved by enhanced training on specific tools, including the mobile phone, to help bring knowledge to these farmers which will result in the better use of technology and support for farmers. If



solutions can be produced, this technology can have the potential to minimize poverty and maximize food security [34].

### *2.5. Lack of Infrastructure, Information, and Technology*

A few barriers have been recorded in Chicago, USA, with urban farmers. Some barriers being faced are limited access to land, insufficient infrastructure, lack of research, and lack of skills in urban agriculture [36]. Studies took place to assess challenges with small-scale farmers in Florida, USA. Major challenges included challenges in accessing new information, and challenges associated with the level of agricultural knowledge within the population, which is required to be familiar with local products, food production, and regarding the inputs needed to produce food [11]. Another study shows how low levels of education and literacy, lack of market information, insecure property rights, poor road and communication infrastructure, and lengthy distances to markets contribute to significant transaction costs for farmers in two KwaZulu-Natal communal areas [41]. Another study used factor analysis to find access to infrastructure in rural communities and the results show the concept of a center-periphery pattern of service infrastructure. Infrastructure development policies and programs need to consider rural economic development [40]. Interviews were conducted to assess the challenges small-scale farmers faced in Tanzania. Results show that they are faced with many challenges including a lack of transport infrastructure, food insecurity, and the lack of access to credit and services. The agriculture sector in Tanzania remains poor, despite multiple initiatives. What stood out was that 78% of these farmers had access to mobile phone technology; however, they had little experience with it and these phones were for personal use [21]. There is potential for information technology to improve the livelihood of small-scale farmers. Even though it cannot solve each problem overall, technology can make a huge impact [21]. In Tanzania, small-scale farmers are faced with several challenges. Some are the lack of infrastructure, food insecurity, and the lack of access to credit and services. In Ghana, farmers face scarce natural resources, limited availability of land, and more [35]. It is considered that direct marketing of produce by locals often cannot meet rapidly expanding consumer demand. A study shows that there is a need for transitional food system infrastructure that benefits direct marketing [39]. Challenges arose among small-scale farmers in rural areas. Having access to mobile phone technology can enable farmers to utilize their resources and increase their income. Technology meets the information needs of these urban farmers such as timely agricultural information through information services, training and learning, and consulting. Having access to this technology can also create interaction between farmers, researchers, and extension workers. This has the potential to enable sustainable farming [46].

## **3. Potential Opportunities for Small-Scale Farming Success**

### *3.1. Potential Technological Opportunities for Small Farmers*

Researchers have been discovering new techniques and technologies to benefit farmers. Advanced precision agriculture technologies have yielded higher efficiency, profitability, and sustainability for the farming sector. However, most of these proven technologies developed in the last few decades might not be suitable for small-scale farms. For example, a sprayer with a wide swath equipped with RTK and variable rate control technology is very useful for improving efficiency and minimizing skips and doubles in large-scale farming, but it is not the right fit economically or practically for a small farm. On the other hand, some of the advanced technologies can be very useful for small farms if these are developed and applied in a specific way. This section (a) lists emerging farm technologies (like agricultural robotics, yield monitoring, automation) and other modern technologies (like artificial intelligence, produce traceability, and smartphone applications); (b) reviews their current status; and (c) provides potential opportunities on how those technologies can be suitable for small-scale farming systems. A summary of these technological advancements relevant to small-scale farming is also listed in Table 2.

**Table 2.** Technological advancements that can benefit small-scale farming systems.

| Technological Advancements                     | System Information                                                                                                               | Relevant Agricultural Operation and Production Stage                                                                                                                                       | Opportunities for Small-Scale Farmers                                                                                                                 |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Unmanned aerial vehicles (UAVs)                | Aerial platforms equipped with sensors and/or crop input delivery mechanisms                                                     | In-season crop sensing, crop care, yield monitoring, and estimation.                                                                                                                       | Weed/pest detection, chemical application, small-scale drone pilot or drone service business, access UAV services as a group of small farmers         |
| Smart sensors and the internet of things (IoT) | Wi-Fi-enabled crop, soil, or weather sensors and data transfer systems                                                           | Crop sensing, soil moisture sensing, weather stations, data transfer for cloud computing                                                                                                   | Inexpensive sensors and IoT technology for notifications and alerts for managing farms, greenhouses, and high tunnels                                 |
| Robotics and artificial intelligence (AI)      | Unmanned ground platforms equipped with sensors and actuators with AI capabilities to process data                               | Fruit/crop harvest, soil/plant tissue sampling, soil preparation, seeding/planting, fertilizer/chemical application, weed management. The potential of fully autonomous farming operations | Inexpensive robotic systems that perform multiple operations on a farm, especially in a greenhouse environment. Labor cost-cutting.                   |
| Yield monitoring and food traceability         | Systems to monitor and measure yield and track post-harvest activities                                                           | Yield monitoring, yield maps, and food traceability especially for organic farmers                                                                                                         | Yield monitoring for budget tracking, payroll management based on fruit picking, and higher sale prices by grocery stores in case food is traceable.  |
| Irrigation automation                          | Systems to actuate irrigation systems automatically or remotely and/or to predict how much and how often farmers should irrigate | Irrigation scheduling and management, crop water stress prediction, remote valve actuation, monitoring the amount of irrigation water used                                                 | Save labor costs and save irrigation water expenses, higher productivity, and opportunity to grow diverse crops                                       |
| Smartphone applications                        | Smartphone-enabled agricultural applications                                                                                     | Farm data management system applications, agricultural vehicle tracking and managing, market updates, weather updates                                                                      | Image data collection, smartphone AI applications for weed/pest/disease identification especially for beginning farmers, market, and weather updates. |

### 3.1.1. Unmanned Aerial Vehicles (UAVs)

UAVs or drones are becoming a mainstream technology on modern farms. According to a market research report by IndustryARC™ (Hyderabad, India), the forecast for the agriculture UAV market is to reach approximately \$6 billion by 2026. There is no doubt that drones provide invaluable data by remotely sensing crops for nutrient management, biotic, or abiotic stress detection, and other aspects of farm management. There are several uses of UAVs in agriculture based on size, payload, and type of sensors mounted on the UAV. Drones offer an effective approach to efficient agricultural management that helps farmers and service providers quickly gain insights into their crops or assets. These provide solutions like scouting and crop monitoring for weed pressure, seed emergence, field conditions, and detecting irrigation system defects [47]. Sophisticated sensors mounted on UAVs are also used for obtaining high-resolution data for gaining insights about soil properties, in-season crop nitrogen management, and early disease detection [48]. Many companies provide services for drone-based applications for farmers, which allow farmers to obtain the imagery and information extracted from the images without investing in in-house drone hardware, training, remote pilot licenses, and image processing software. As prices decrease for drone technology, drones may be used for small-scale operations as well [49]. Some studies show successful implementation of drone technology and satellite

imagery for small-scale farms. A study evaluated spatial resolutions using drone imagery to correlate with the crop status on smallholder farms. According to the results, most correlated signals for crop health were nearer the maize plant scale (i.e., 14–27 cm) [50].

Short battery life has been the biggest challenge for agriculture drones. This limitation makes a compelling case for the use of drones for small-scale operations. For small-scale operations that have multiple crops in a small area, drones have the potential to offer variable rate application of chemicals and fertilizers. However, there is a need for on-farm studies to validate the effectiveness and efficacy of using UAVs for dry, liquid, or seed product applications. In fact, drones are becoming a popular option for chemical application in some Asian countries where the average farm size is relatively smaller. There is an opportunity for small-scale farmers to be part of agricultural cooperatives to gain access to otherwise costly drone-based applications for agriculture. For example, imagery services provided by drones for weed and other crop stress detection might not be a feasible option for individual growers. However, this might be a practical and suitable use case for small-scale operations as part of a bigger group of farmers.

### 3.1.2. Smart Sensors and the Internet of Things (IoT)

Sensors are not at all new to the agricultural industry. Data obtained using sensors from soil, crop, machine, and environment plays a key role in the implementation of precision agriculture techniques at the farm. Recently, agriculture sensors have been becoming smarter and more robust. Handheld, tractor- or implement-mounted, drone-mounted, weather stations, under-the-soil sensors, on-the-go soil and crop sensors are some usual places and methods to acquire valuable data for precision agriculture implementation. Improvement in wireless technologies like LoRa, Bluetooth, and cellular service coverage at farms allows wireless networks and smart sensors to continuously send data to the cloud services. Consequently, this continuous flow of data combined with cloud computing has made it possible for the agriculture industry to bring various IoT solutions to market. This data can be accessed in real time by various mobile phone applications in a very useful and presentable form which allows growers to make wise decisions and keep track of their farm and market activities at their fingertips. Technologies like smart sensors, and the internet of things (IoT) are much needed for small-scale specialty growers and urban agriculture growers [51]. A lot of time and labor are required for the record-keeping required to run small-scale specialty crop farm operations. Inexpensive sensors and IoT solutions in combination with mobile apps can be very helpful in developing affordable and easy-to-use solutions to ensure high-quality data collection and record-keeping throughout the growing season.

Greenhouse, hydroponics, and aquaponic systems claim to be more efficient and sustainable ways to produce food by using less inputs (water, pesticides, fertilizers). Adoption of these systems is on the rise in many urban centers as it is taken up by small and first-time growers [52]. It addresses food security in many areas and is a desirable choice for many small-scale and beginning farmers. The major challenge is to ensure profitability for small-scale specialty crop growers and urban agriculture operations. Small-scale growers manually record, monitor, and control a lot of variables to run their operations on organic farms, family farms, greenhouses, hydroponic, or aquaponic systems [53]. Affordable and reliable sensors can be useful in these systems. The concept of precision agriculture must be adopted by small-scale urban growers by using smart sensing, control, and IoT solutions for automatically and precisely monitoring the processes to run their operations successfully and profitably.

### 3.1.3. Robotics and Artificial Intelligence (AI)

Data-driven, and AI-based digital agriculture has the potential to make agriculture more efficient and profitable [54]. It can provide radical solutions to challenges in agricultural production systems not just by automating data acquisition and processing, but also by automating complex, repetitive, and labor-intensive agricultural operations like weed

management, pest management, disease management, crop scouting, fruit picking, and harvest automation [55]. Agriculture robots are considered the most important solution to solving the prolonged issue of labor shortages in different agricultural communities around the world. Labor shortages in agriculture are a common problem faced by both large-scale and small-scale operations. Robotic platforms are also going to play a crucial role in meeting ever-increasing global demand for food. Almost all the big agriculture equipment manufacturing companies are investing to deliver autonomous solutions to the market as early as possible. For example, Raven Industries, recently acquired by CNH Industrial has launched a fully autonomous platform branded as OMNiPOWER which is capable of interchanging implements for handling spraying or spreader operations suited for large-scale row crop production fields.

The smaller size of most of these robotic platforms as compared to traditional tractors and implements brings great opportunity for the use of these technologies by small-scale growers and for specialty crops. One major challenge is that the cost of robotic systems equipped with AI solutions is often not affordable for small-scale farmers. To reach a wide range of farmers, especially smallholders and new farmers, the agriculture robots need to have a cost-effective, simple, and easy-to-use user interface and must be suitable for performing field operations in various types of field conditions and terrains. Another challenge is that current robotic and AI-based precision agriculture solutions have also been divided into different sub-systems including robots, drones, tractors, and various other mechanical equipment. Little effort has been made to integrate these sub-systems to make one affordable and coherently integrated system. These sub-systems need to be integrated in a way that ensures affordability and convenience to the end-user.

#### 3.1.4. Yield Monitoring and Food Traceability

Small-scale farmers often have small margins and most farm operations are handled manually, which requires a lot of manual labor hours. Ref. [56] developed and evaluated a cloud-based harvest management information system (HMIS) for orchard crops that provides real-time access to harvest data, and it can also provide yield maps, payroll, labor productivity records, and real-time decision-making and fruit-handling logistics. This type of specialty crop harvest management system can be very useful for many specialty crop types and for various other farm operations. Automation of book-keeping, scheduling, managing farm operations, and yield monitoring in one central cloud-based app not only makes small-scale farming efficient and profitable, but it can also provide key data-driven decision-making statistics. These apps and online portals need to be customizable by the farmers based on the crop type, and farm size. These systems can also play a very important role in food traceability, which could add a lot of value to small-scale farm operations. Consumers demand more information about the origin and overall journey of the food they consume. Based on food safety regulations and consumer awareness, most fresh produce retail and grocery stores have started to require food traceability. This plays a crucial role in taking quick action in case of food recalls. Because there is a possibility of contamination in the distribution channel, and these incidents can cause unbearable losses to an industry worth billions of dollars, these new technologies are in high demand. Low-cost GPS receivers, RFID, and QR code methods can be used to gather harvest data from small manual operations. This data becomes the key asset and a starting point for the farm-to-fork journey for supply chain transparency and food traceability.

#### 3.1.5. Irrigation Automation

Irrigation automation or precision irrigation (PI) emerged as the most promising technique for farmers in irrigated agriculture in arid and semi-arid regions where water is scarce. Techniques like the development of irrigation management zones, variable rate irrigation (VRI), and plant water stress sensing have garnered a lot of attention from researchers as well as from growers. There is no doubt that demand for automated irrigation systems that integrate irrigation requirements, soil moisture sensing, rainfall data, and

IoT technology is on the rise in arid and semi-arid regions, irrespective of farm size [57]. On the other hand, agriculture in temperate and continental climates usually has a lot of non-irrigated acreage. Considering the traditional row crops in the Midwestern US, only 47% and 64% of total harvested soybean and corn crops were under irrigation in 2013 [58]. Irrigated acreage numbers do not seem very high; however, these numbers have been on the rise recently and it should be noted that irrigation plays a critical role for small-scale specialty crop farming, even in the Midwestern US.

There is a big opportunity for irrigation automation for small-scale operations as well. Most of the work on small-scale specialty crop farming operations is performed manually. Weeding and harvesting are the most labor-intensive operations, and it is quite expensive and complex to automate these operations. To save labor costs, small-scale specialty crop operations will certainly benefit from irrigation automation for farm operations based on soil moisture sensing. This information can be combined with soil type, field slope, and rainfall data to implement an automated irrigation control setup. Future climate change effects in humid subtropical regions are predicting possible negative effects on maize yields. Most of agriculture is rain-dependent in these areas; therefore, this demands better preparation and adaptation strategies [59]. Specialty crop production with irrigation systems can also play an important role in crop diversification for these climates. Because unpredictable water supplies for specialty and high-value crops can be a critical risk, this is one of the primary reasons that hinder crop diversification in humid climates.

#### 3.1.6. Use of Smartphones for Agricultural Applications

Cellphones and smart mobile phones are becoming the most important tools of communication for farmers to access agricultural-related information. As smartphones have reached the hands of most middle-class and small farmers' hands, cellular data-enabled services provide ways to communicate information to the knowledge and data-intensive agriculture sector [60]. Smartphones are also helping farmers in sharing their experience, data, and information by providing easy access to the internet and mobile applications. A survey was conducted to evaluate the adoption of smartphone and cellular services by Kenyan farmers related to price and activity alerts for agriculture commodities [61]. Smartphones can play a key role for businesses entering rural markets which are lagging in gaining access to advanced agricultural techniques. In Africa, there has been a lack of information leading to harvest losses among small-scale growers. It was stated that the use of mobile phones may be beneficial to farmers. This will help by connecting farmers and closing the information gap [62].

Smartphones not only provide easy access to weather information and market information; they also remove the technological barrier and provide easy-to-use applications for operating hardware, accessing sensor data, real-time IoT solutions and cloud services, and farm management applications with affordable prices. These applications and websites can help small-scale farmers in communication, access to information, tracking expenses and yield, managing farms, and eventually that can lead to better decision making, and increases in productivity and profitability.

#### 3.2. Other Considerations for the Success of Smallholders

No doubt technological advancements are the key to success and sustainability of small-scale farming in the future. There are many other considerations that communities, policymakers, funding agencies, private companies, and higher education and research institutes need to think about in terms of contributing to the success of small-scale farming. Many government organizations have launched several initiatives, like microloan programs launched by the United States Department of Agriculture (USDA) to mitigate the economic challenges of small-scale farmers, beginning farmers, and non-traditional farm operations in the past [63]. More initiatives should be launched for more crops at regional scales for small farmers. Higher education institutes need to encourage research programs and university-level courses that focus on educating and developing solutions tailored for



small-scaling farming. Private industry also has a potential opportunity to collaborate with universities and funding agencies to develop tools, equipment, and precision agricultural solutions for small-scale farming and specialty crops.

#### 4. Conclusions

Small-scale farms play a key role in producing fresh produce in developed countries like the USA and small-scale farming is a primary source of food in many developing countries. Advanced agricultural technologies have been mostly out of reach from small-scale farmers. This article discussed the importance of small-scale farming, listed agricultural technologies relevant to small-scale farming, and provided potential opportunities on how these technologies can be useful for small-scale farming. In conclusion, UAV technology has shown promise and has the potential to be used for liquid and dry product applications, which can be huge for small-scale farms. Irrigation automation for small farms can help in higher productivity and it also solves the challenge of labor shortages for many other manual operations on small-scale farms. Robotic solutions can solve the labor issue for small-scale farms; however, cost remains a barrier for robotics to be a viable solution for small-scale farmers. A combination of smart sensors, IoT solutions, and smartphone applications has huge potential for small-scale farmers for data-driven decision making, automation, bookkeeping, yield tracking, food traceability, and cloud-based applications for managing farm operations.

**Author Contributions:** Conceptualization, R.D.; methodology, R.D.; formal analysis, Q.M. and R.D.; data curation, Q.M.; writing—original draft preparation, Q.M.; writing—review and editing, R.D.; supervision, R.D.; project administration, R.D.; funding acquisition, R.D. All authors have read and agreed to the published version of the manuscript.

**Funding:** This research was funded by the USDA NIFA Evans-Allen program: Advanced Agriculture Technologies for Small Scale Farms, 2022–2027.

**Acknowledgments:** This research was supported by the USDA NIFA Evans-Allen program: Advanced Agriculture Technologies for Small Scale Farms, 2022–2027.

**Conflicts of Interest:** The authors declare no conflict of interest. The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.

#### References

- Lowder, S.K.; Sánchez, M.V.; Bertini, R. Which farms feed the world and has farmland become more concentrated? *World Dev.* **2021**, *142*, 105455. [CrossRef]
- Ricciardi, V.; Ramankutty, N.; Mehrabi, Z.; Jarvis, L.; Chookolingo, B. How much of the world's food do smallholders produce? *Glob. Food Secur.* **2018**, *17*, 64–72. [CrossRef]
- USDA. *Family Farms Are the Focus of New Agriculture Census Data*; USDA: Washington, DC, USA, 2017; No. 0066.15. [CrossRef]
- Bui, T.N.; Nguyen, A.H.; Le, T.T.; Nguyen, V.P.; Le, T.T.; Tran, T.T.; Nguyen, N.M.; Le, T.K.; Nguyen, T.K.; Nguyen, T.T.; et al. Can a short food supply chain create sustainable benefits for small farmers in developing countries? An exploratory study of Vietnam. *Sustainability* **2021**, *13*, 2443. [CrossRef]
- Cui, Z.; Zhang, H.; Chen, X.; Zhang, C.; Ma, W.; Huang, C.; Zhang, W.; Mi, G.; Miao, Y.; Li, X.; et al. Pursuing sustainable productivity with millions of smallholder farmers. *Nature* **2018**, *555*, 363–366. [CrossRef]
- Jouzi, Z.; Azadi, H.; Taheri, F.; Zarafshani, K.; Gebrehiwot, K.; Van Passel, S.; Lebailly, P. Organic farming and small-scale farmers: Main opportunities and challenges. *Ecol. Econ.* **2017**, *132*, 144–154. [CrossRef]
- Schimmelpfennig, D.; Ebel, R. On the Doorstep of the Information Age: Recent Adoption of Precision Agriculture. Economic Research Service 2011, Paper No. EIB-80. 2011. Available online: <https://ssrn.com/abstract=2692052> (accessed on 2 August 2023).
- Batte, M.T.; Arnholt, M.W. Precision farming adoption and use in Ohio: Case studies of six leading-edge adopters. *Comput. Electron. Agric.* **2003**, *38*, 125–139. [CrossRef]
- Mondal, P.; Basu, M. Adoption of precision agriculture technologies in India and in some developing countries: Scope, present status and strategies. *Prog. Nat. Sci.* **2009**, *19*, 659–666. [CrossRef]
- Chen, H.; Ellett, J.K.; Phillips, R.; Feng, Y. Small-scale produce growers' barriers and motivators to value-added business: Food safety and beyond. *Food Control* **2021**, *130*, 108192. [CrossRef]

11. Mpandeli, S.; Maponya, P. Constraints and challenges facing the small-scale farmers in Limpopo Province, South Africa. *J. Agric. Sci.* **2014**, *6*, 135. [CrossRef]
12. Stringer, L.C.; Twyman, C.; Gibbs, L.M. Learning from the South: Common challenges and solutions for small-scale farming. *Geogr. J.* **2008**, *174*, 235–250. [CrossRef]
13. Goodwin, J.N.; Gouldthorpe, J.L. Small farmers, big challenges: A needs assessment of Florida small-scale farmers' production challenges and Training needs. *J. Rural. Soc. Sci.* **2013**, *28*, 3.
14. Chisita, C.T. An investigation into the use of ICT in the provision of agricultural information to small scale farmers in Harare. In Proceedings of the World Library and Information Congress: 76th IFLA General Conference and Assembly, Gothenburg, Sweden, 25 June 2010; pp. 10–15. Available online: <https://cdn.ifla.org/past-wlic/2010/85-chisita-en.pdf> (accessed on 3 August 2023).
15. Gebremedhin, T.G.; Christy, R.D. Structural changes in US agriculture: Implications for small farms. *J. Agric. Appl. Econ.* **1996**, *28*, 57–66. [CrossRef]
16. Yaro, J.A. The perception of and adaptation to climate variability/change in Ghana by small-scale and commercial farmers. *Reg. Environ. Change* **2013**, *13*, 1259–1272. [CrossRef]
17. Bezus, R.; Samofal, O. Challenges of small-scale farming in Ukraine. *AgroLife Sci. J.* **2019**, *8*, 35–42.
18. Kang'ethe, S.M.; Serima, J. Exploring challenges and opportunities embedded in small-scale farming in Zimbabwe. *J. Hum. Ecol.* **2014**, *46*, 177–185. [CrossRef]
19. Boys, K.A.; Fraser, A.M. Linking small fruit and vegetable farmers and institutional foodservice operations: Marketing challenges and considerations. *Renew. Agric. Food Syst.* **2019**, *34*, 226–238. [CrossRef]
20. Ugulumu, E.S.; Inanga, E.L. Market Access and Sunflower Marketing: Challenges and Prospects to Small Scale Farmers in Tanzania. 2014. Available online: <https://hdl.handle.net/20.500.12018/7385> (accessed on 3 August 2023).
21. Misaki, E.; Apiola, M.; Gaiani, S. Technology for small scale farmers in Tanzania: A design science research approach. *Electron. J. Inf. Syst. Dev. Ctries.* **2016**, *74*, 1–15. [CrossRef]
22. Thibane, Z.; Soni, S.; Phali, L.; Mdoda, L. Factors impacting sugarcane production by small-scale farmers in KwaZulu-Natal Province-South Africa. *Heliyon* **2023**, *9*, e13061. [CrossRef]
23. Cairns, J.E.; Chamberlin, J.; Rutsaert, P.; Voss, R.C.; Ndhlela, T.; Magorokosho, C. Challenges for sustainable maize production of smallholder farmers in sub-Saharan Africa. *J. Cereal Sci.* **2021**, *101*, 103274. [CrossRef]
24. Ncoyini, Z.; Savage, M.J.; Strydom, S. Limited access and use of climate information by small-scale sugarcane farmers in South Africa: A case study. *Clim. Serv.* **2022**, *26*, 100285. [CrossRef]
25. Wortman, S.E.; Lovell, S.T. Environmental challenges threatening the growth of urban agriculture in the United States. *J. Environ. Qual.* **2013**, *42*, 1283–1294. [CrossRef]
26. Mbuli, C.S.; Fonjong, L.N.; Fletcher, A.J. Climate change and small farmers' vulnerability to food insecurity in Cameroon. *Sustainability* **2021**, *13*, 1523. [CrossRef]
27. Hoffmann, U. Assuring food security in developing countries under the challenges of climate change: Key trade and development issues of a fundamental transformation of agriculture. In *Discussion Papers-United Nations Conference on Trade and Development (UNCTAD)*; No. 201; United Nations Conference on Trade and Development: Geneva, Switzerland, 2011. Available online: <https://econpapers.repec.org/paper/uncdisap/201.htm> (accessed on 2 August 2023).
28. Chen, H.; Benjamin, T.; Guan, W.; Feng, Y. Food Safety Education Needs Assessment for Small-Scale Produce Growers Interested in Value-Added Food Production. *J. Food Prot.* **2022**, *85*, 220–230. [CrossRef] [PubMed]
29. Olayemi, F.F.; Adegbola, A.; Bamishaiye, E.I. Assessment of post-harvest challenges of small-scale farm holders of tomatoes, bell and hot pepper in some local government areas of Kano State, Nigeria. *Bayero J. Pure Appl. Sci.* **2010**, *3*, 39–42. [CrossRef]
30. Afenyo, J.S. Making Small Scale Farming Work in Sub-Saharan Africa. Global Forum on Food Security and Nutrition FSN Forum, FAO. 2012. Available online: [https://www.fao.org/fileadmin/user\\_upload/fsn/docs/Making\\_Small\\_Scale\\_Farming\\_Work\\_in\\_SSA\\_-\\_By\\_Joy\\_S.\\_Afenyo.pdf](https://www.fao.org/fileadmin/user_upload/fsn/docs/Making_Small_Scale_Farming_Work_in_SSA_-_By_Joy_S._Afenyo.pdf) (accessed on 12 August 2023).
31. Mavhunduse, F.; Holmner, M.A. Utilisation of Mobile Phones in Accessing Agricultural Information by Smallholder Farmers in Dzindi Irrigation Scheme in South Africa. 2019. Available online: <https://hdl.handle.net/2263/76997> (accessed on 11 August 2023).
32. Zaca, F. Social Grant Dependence, Irrigation Water Use and On-Farm Entrepreneurial Spirit: A Behavioural Explanation for Smallholders in KwaZulu-Natal. Ph.D. Diss. 2018. Available online: <https://ukzn-dspace.ukzn.ac.za/handle/10413/17182> (accessed on 7 August 2023).
33. Mugwisi, T.; Mostert, J.; Ocholla, D.N. Access to and utilization of information and communication technologies by agricultural researchers and extension workers in Zimbabwe. *Inf. Technol. Dev.* **2015**, *21*, 67–84. [CrossRef]
34. Misaki, E.; Apiola, M.; Gaiani, S.; Tedre, M. Challenges facing sub-Saharan small-scale farmers in accessing farming information through mobile phones: A systematic literature review. *Electron. J. Inf. Syst. Dev. Ctries.* **2018**, *84*, e12034. [CrossRef]
35. Elijah, O.; Rahman, T.A.; Orikumhi, I.; Leow, C.Y.; Hindia, M.N. An overview of Internet of Things (IoT) and data analytics in agriculture: Benefits and challenges. *IEEE Internet Things J.* **2018**, *5*, 3758–3773. [CrossRef]
36. Castillo, S.R.; Winkle, C.R.; Krauss, S.; Turkewitz, A.; Silva, C.; Heinemann, E.S. Regulatory and other barriers to urban and peri-urban agriculture: A case study of urban planners and urban farmers from the greater Chicago metropolitan area. *J. Agric. Food Syst. Community Dev.* **2013**, *3*, 155–166. [CrossRef]
37. Kalusopa, T. The challenges of utilizing information communication technologies (ICTs) for the small-scale farmers in Zambia. *Libr. Hi Tech.* **2005**, *23*, 414–424. Available online: <http://168.167.8.130/handle/10311/764> (accessed on 15 August 2023). [CrossRef]

38. Getahun, A.A. Challenges and opportunities of information and communication technologies for dissemination of agricultural information in Ethiopia. *Int. J. Agric. Ext.* **2020**, *8*, 57–65. Available online: <https://esciencepress.net/journals/index.php/IJAE/article/view/3069> (accessed on 10 August 2023). [CrossRef]
39. Bloom, J.D.; Hinrichs, C.C. Moving local food through conventional food system infrastructure: Value chain framework comparisons and insights. *Renew. Agric. Food Syst.* **2011**, *26*, 13–23. [CrossRef]
40. Makhura, M.N.; Wasike, W.S. Patterns of access to rural service infrastructure: The case of farming households in Limpopo Province. *Agrekon* **2003**, *42*, 129–143. [CrossRef]
41. Ortmann, G.F.; King, R.P. Agricultural cooperatives II: Can they facilitate access of small-scale farmers in South Africa to input and product markets? *Agrekon* **2007**, *46*, 219–244. [CrossRef]
42. Bisaga, I.; Parikh, P.; Loggia, C. Challenges and opportunities for sustainable urban farming in South African low-income settlements: A case study in Durban. *Sustainability* **2019**, *11*, 5660. [CrossRef]
43. Muema, E.; Mburu, J.; Coulbaly, J.; Mutune, J. Determinants of access and utilisation of seasonal climate information services among smallholder farmers in Makueni County, Kenya. *Heliyon* **2018**, *4*, e00889. [CrossRef]
44. Barik, A.K. Organic Farming in India: Present Status, Challenges and Technological Breakthrough. In Proceedings of the 3rd Conference on Bio-Resource and Stress Management International, Jaipur, India, 8–11 November 2017; pp. 101–110.
45. Mohammad, Z.H.; Yu, H.; Neal, J.A.; Gibson, K.E.; Sirsat, S.A. Food safety challenges and barriers in southern United States farmers markets. *Foods* **2019**, *9*, 12. [CrossRef] [PubMed]
46. Balogun, A.L.; Adebisi, N.; Abubakar, I.R.; Dano, U.L.; Tella, A. Digitalization for transformative urbanization, climate change adaptation, and sustainable farming in Africa: Trend, opportunities, and challenges. *J. Integr. Environ. Sci.* **2022**, *19*, 17–37. [CrossRef]
47. Tripicchio, P.; Satler, M.; Dabisias, G.; Ruffaldi, E.; Avizzano, C.A. Towards smart farming and sustainable agriculture with drones. In Proceedings of the 2015 International Conference on Intelligent Environments, Prague, Czech Republic, 15–17 July 2015; pp. 140–143. [CrossRef]
48. Malveaux, C.; Hall, S.G.; Price, R. Using drones in agriculture: Unmanned aerial systems for agricultural remote sensing applications. In *2014 Montreal, Quebec Canada July 13–July 16, 2014*; American Society of Agricultural and Biological Engineers: St. Joseph, MI, USA, 2014; p. 1. [CrossRef]
49. Ehsani, R. Precision Agriculture for small growers. *Resour. Mag.* **2011**, *18*, 1.
50. Peter, B.G.; Messina, J.P.; Carroll, J.W.; Zhi, J.; Chimonyo, V.; Lin, S.; Snapp, S.S. Multi-Spatial Resolution Satellite and sUAS Imagery for Precision Agriculture on Smallholder Farms in Malawi. *Photogramm. Eng. Remote Sens.* **2020**, *86*, 107–119. [CrossRef]
51. Gyltshen, P.; Osathanunkul, K. Linking Small-Scale Farmers to Market Using ICT. In Proceedings of the 2018 International Conference on Digital Arts, Media and Technology, Phayao, Thailand, 25–28 February 2018; pp. 120–125. [CrossRef]
52. Han, G.; Schoolman, E.D.; Morton, L.W.; Arbuckle, J.G. 2017 Survey of Specialty Crop Growers in Michigan and Ohio. 2018. Available online: <https://www.climatehubs.usda.gov/sites/default/files/2017%20Survey%20of%20Specialty%20Crop%20Growers%20in%20Michigan%20and%20Ohio.pdf> (accessed on 7 August 2023).
53. Lundqvist, P. Ergonomics in Organic Farming. In Proceedings of the Human Factors and Ergonomics Society Annual Meeting 2000, San Diego, CA, USA, 30 July–4 August 2000; Sage: Los Angeles, CA, USA; Volume 44, No. 22. pp. 655–657. [CrossRef]
54. Torero, M. Robotics and AI in Food Security and Innovation: Why They Matter and How to Harness Their Power. In *Robotics, AI, and Humanity: Science, Ethics, and Policy*; Springer: Cham, Switzerland, 2021; pp. 99–107. [CrossRef]
55. Marinoudi, V.; Sørensen, C.G.; Pearson, S.; Bochtis, D. Robotics and labour in agriculture. A context consideration. *Biosyst. Eng.* **2019**, *184*, 111–121. [CrossRef]
56. Ampatzidis, Y.; Tan, L.; Haley, R.; Whiting, M.D. Cloud-based harvest management information system for hand-harvested specialty crops. *Comput. Electron. Agric.* **2016**, *122*, 161–167. [CrossRef]
57. Lichtenberg, E.; Majsztrik, J.; Saavoss, M. Grower demand for sensor-controlled irrigation. *Water Resour. Res.* **2015**, *51*, 341–358. [CrossRef]
58. USDA. National Agriculture Statistics Service. 2013. Available online: <https://www.nass.usda.gov/> (accessed on 12 May 2023).
59. Hernandez-Ochoa, I.M.; Asseng, S. Cropping systems and climate change in humid subtropical environments. *Agronomy* **2018**, *8*, 19. [CrossRef]
60. Masimba, F.; Appiah, M.; Zuva, T. Influencing Factors of Mobile Technology Adoption by Small Holder Farmers in Zimbabwe. In *Proceedings of the Computational Methods in Systems and Software 2019*; Springer: Cham, Switzerland; pp. 125–134. [CrossRef]
61. Krell, N.T.; Giroux, S.A.; Guido, Z.; Hannah, C.; Lopus, S.E.; Caylor, K.K.; Evans, T.P. Smallholder farmers' use of mobile phone services in central Kenya. *Clim. Dev.* **2021**, *13*, 215–227. [CrossRef]
62. Masuka, B.; Matenda, T.; Chipomho, J.; Mapope, N.; Mupeti, S.; Tatsvarei, S.; Ngezimana, W. Mobile phone use by small-scale farmers: A potential to transform production and marketing in Zimbabwe. *S. Afr. J. Agric. Ext.* **2016**, *44*, 121–135. [CrossRef]
63. USDA. Microloan Programs. Available online: <https://www.fsa.usda.gov/programs-and-services/farm-loan-programs/microloans/index> (accessed on 18 August 2023).

**Disclaimer/Publisher's Note:** The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.